



Family Name

Given Name

Student No.

Signature

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

MID-TERM EXAMINATION

Term 3, 2024

ELEC1111

Electrical and Telecommunications Engineering

TIME ALLOWED:	90 minutes (10' reading + 80' working)
TOTAL MARKS:	100
TOTAL NUMBER OF QUESTIONS:	5

THIS EXAM CONTRIBUTES 17.5% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 7 pages.

All questions **MUST** be answered in the corresponding **labelled page(s) in the answer booklet**. Read the first page of the answer booklet carefully and follow all the instructions. Any working in this **exam paper will NOT be marked**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

Formulae you may use:

$$i = \frac{dq}{dt}$$

$$v = \frac{dw}{dq}$$

$$p = \frac{dw}{dt}$$

Voltage division:

$$v_n = v \frac{R_n}{R_{eq}}$$

Current division:

$$i_n = i \frac{R_{eq}}{R_n}$$

Maximum power transfer:

$$R_L = R_{th}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

Capacitors and RC circuits:

$$q = Cv$$

$$i = C \frac{dv}{dt}$$

$$w_c = \frac{1}{2} C v^2$$

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-\frac{(t-t_0)}{\tau}}, \tau = RC$$

QUESTION 1 [16 marks]

a. (9 marks) For the circuit shown in Figure 1,

- (5 marks) Calculate the equivalent resistance R_{eq} seen from terminals $a-b$.
- (2 marks) A 5-V battery is connected to terminals $a-b$. What is the total current flowing through the circuit?
- (2 marks) Keeping the battery connected, calculate the total energy that would be delivered to the equivalent resistance R_{eq} during a period of 2 hours. Express your answer in Joules (J).

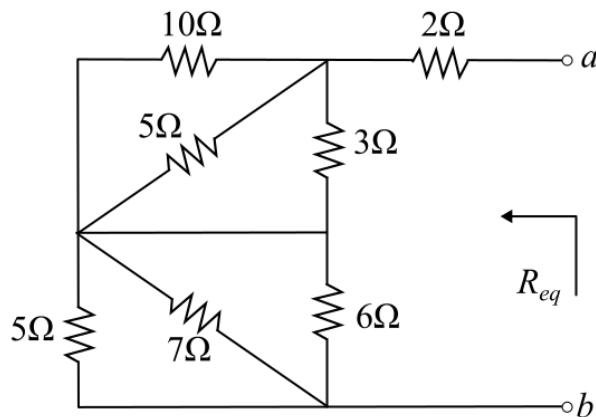


Figure 1

b. (7 marks) A circuit is designed to indicate whether your academic advisor is available for a chat using a traffic light system, as shown in Figure 2. A green LED is used to show availability (lights up when the advisor is free) and a red LED is used to indicate unavailability (lights up when the advisor is busy). The advisor controls the LEDs by controlling the polarity of the battery. The input voltage (V_{in}) is set at 9 V and the forward voltage drop of each LEDs is 1 V.

- (4 marks) Calculate the current flowing through the lights (i_{lights}) in the circuit shown in Figure 2. Is the academic advisor available or unavailable?
- (3 marks) If the advisor swaps the polarity of the battery (i.e., the minus sign is at the top and the plus sign is at the bottom), what is the new current flowing through the lights (i_{lights})? Is the academic advisor available or unavailable now?

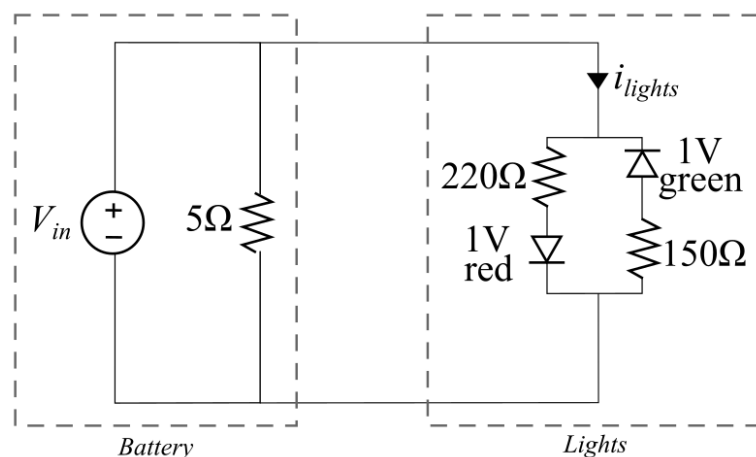


Figure 2

QUESTION 2 [22 marks]

- a. (12 marks) For the circuit provided in Figure 3,
- (9 marks) Use nodal analysis to calculate V_1 , V_2 and V_3 .
 - (1 mark) Using your result from i), calculate i_0 .
 - (2 marks) Using your result from i), calculate the power in resistor R_L .

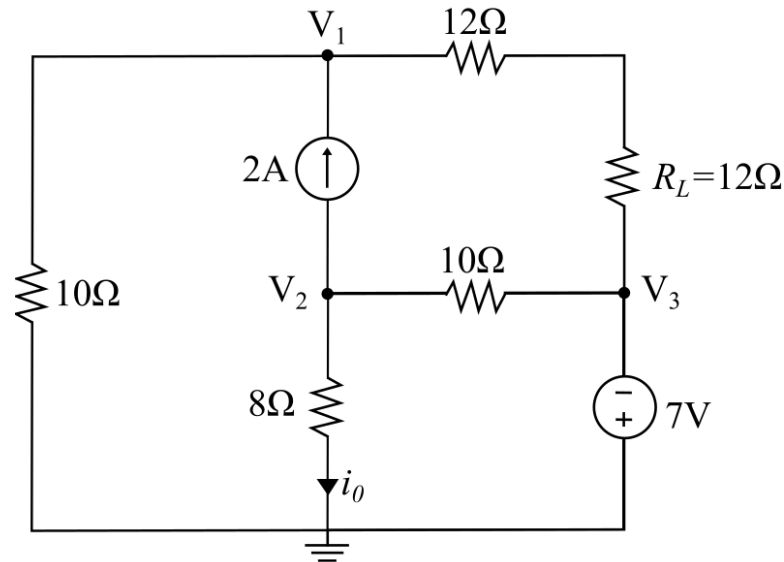


Figure 3

- b. (10 marks) For the circuit shown in Figure 4, apply mesh analysis and write down the mesh equations using the labels provided.

Note: Simplify the equations in the form $aI_1 + bI_2 + cI_3 = d$, where coefficients a , b , c and d can be positive or negative numbers, but DO NOT solve them.

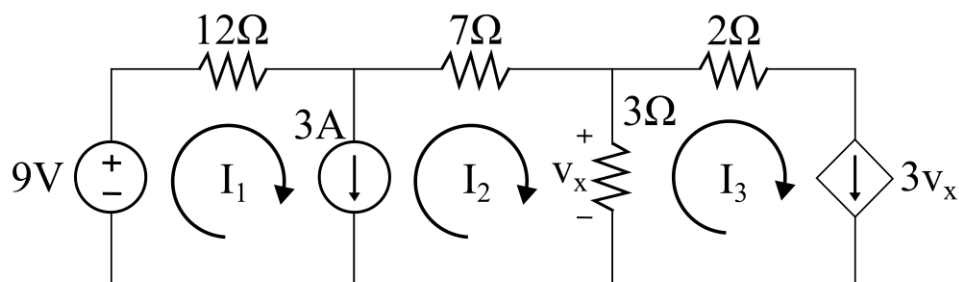


Figure 4

QUESTION 3 [16 marks]

The circuit in Figure 5 shows a solar-powered garden light. The solar panel charges the battery. The battery is then used to turn the light ON when triggered by a motion sensor. For simplicity, the motion sensor is modelled as a Switch and the lights are modelled as a 10Ω resistor.

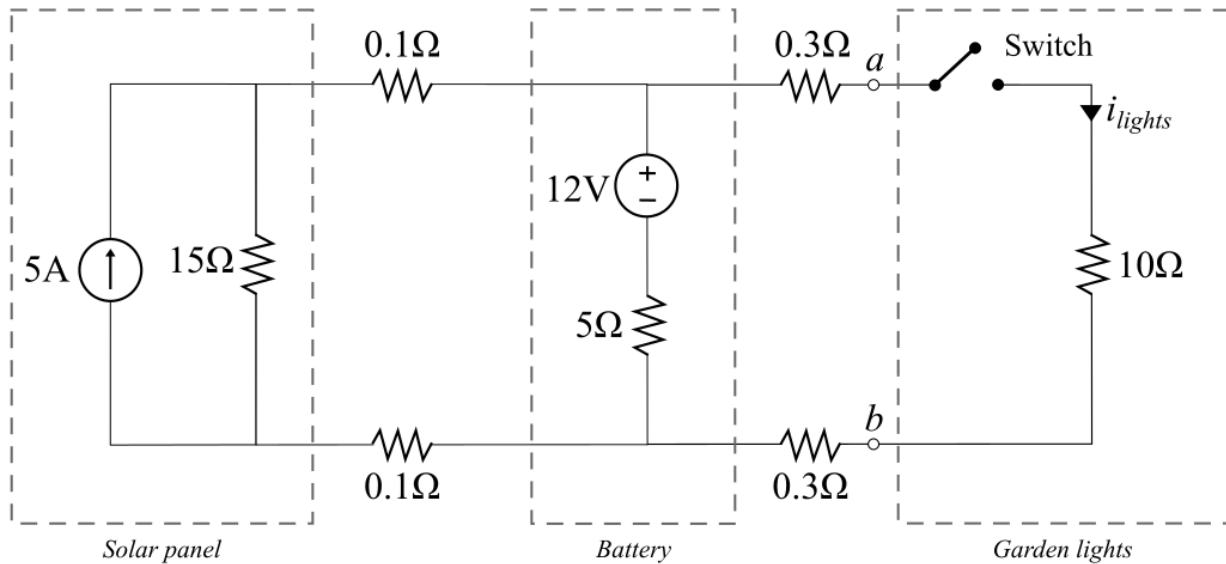


Figure 5

- i. **(15 marks)** When motion is detected (i.e., the switch is closed), calculate the current i_{lights} using superposition.
- ii. **(1 mark)** Find the power in the garden lights, denoted by the 10Ω resistor.

QUESTION 4 [29 marks]

- a. **(12 marks)** For the circuit provided in Figure 6, use a succession of source transformations (only source transformations and resistor/source simplification) to find an equivalent circuit consisting of a single voltage source and single resistor as seen from terminals a - b . Then determine the Thévenin voltage (V_{Th}) and Thévenin resistance (R_{Th}) at the terminals a - b .

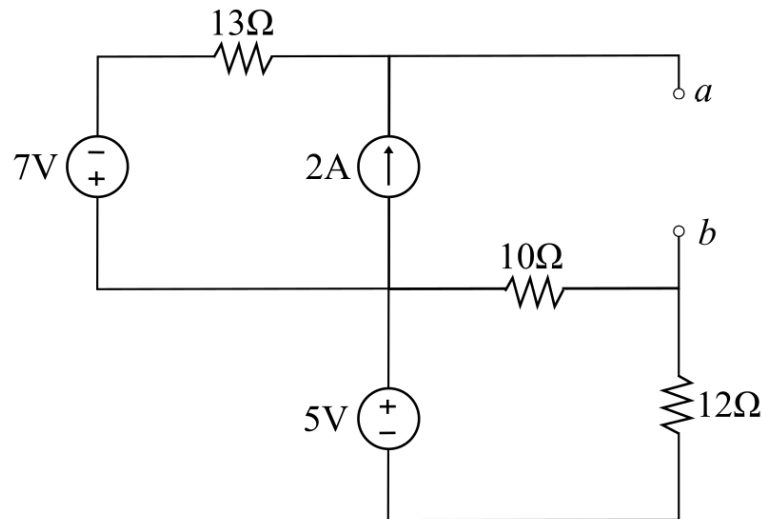


Figure 6

- b. **(17 marks)** The circuit shown in Figure 7 is being used to power a toy-airplane propeller. The propeller can be modelled as a single resistor connected to terminals a - b , as shown in Figure 7.
- (8 marks)** Calculate the Thévenin voltage (V_{Th}) at terminals a - b .
 - (7 marks)** Calculate the value of the propeller resistance that will ensure maximum transfer of power from the circuit to the propeller.
 - (2 marks)** Find the maximum power that can be delivered to the propeller.

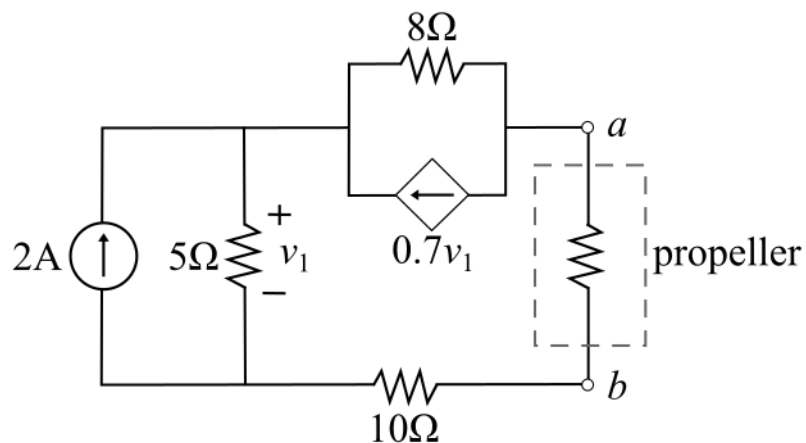


Figure 7

QUESTION 5 [17 marks]

There is no energy stored in the capacitor in the circuit shown in Figure 8 when switch 1 closes at $t = 0$ s. The capacitance C is given as 500 mF. Calculate:

- i. (7 marks) $v_0(t)$ for $0 < t \leq 1800$ s.
- ii. (8 marks) $v_0(t)$ for $t > 1800$ s.
- iii. (2 marks) $i_0(t)$ for $t > 1800$ s.

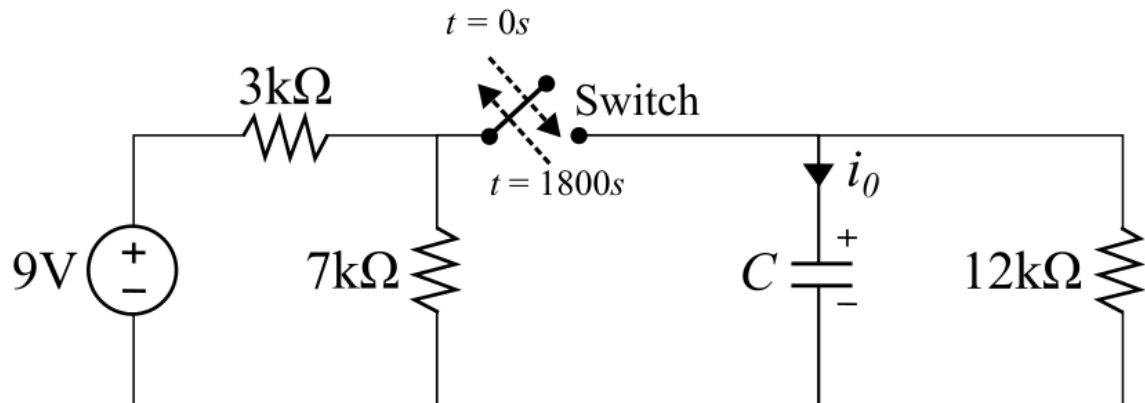


Figure 8